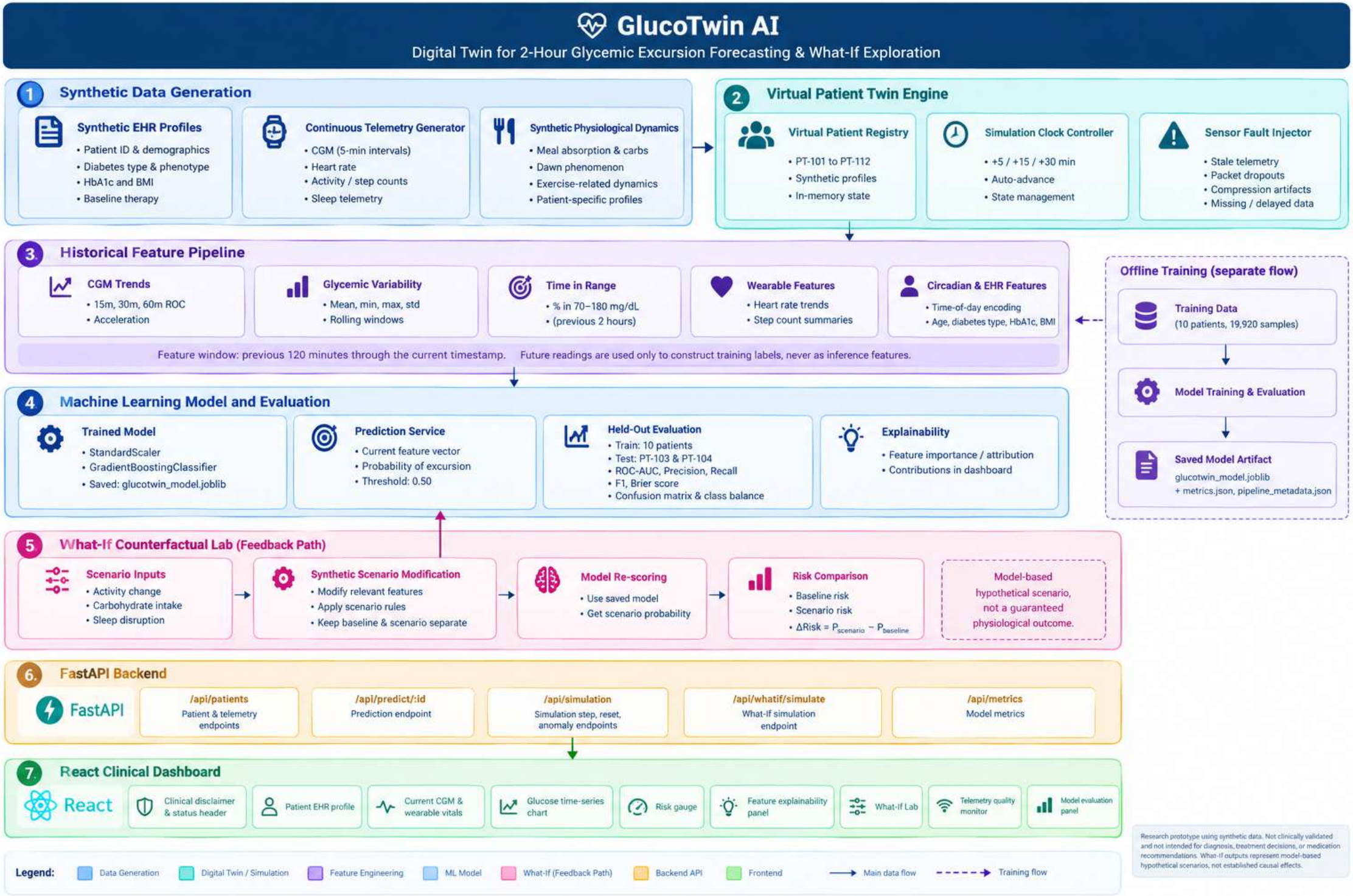

GlucoTwin AI
Digital Twin for 2-Hour Glycemic Excursion Forecasting & What-If Exploration
1 Synthetic Data Generation
Synthetic EHR Profiles
• Patient ID & demographics
• Diabetes type & phenotype
• HbA1c and BMI
• Baseline therapy
Continuous Telemetry Generator
• CGM (5-min intervals)
• Heart rate
• Activity / step counts
• Sleep telemetry
Synthetic Physiological Dynamics
• Meal absorption & carbs
• Dawn phenomenon
• Exercise-related dynamics
• Patient-specific profiles
2 Virtual Patient Twin Engine
Virtual Patient Registry
• PT-101 to PT-112
• Synthetic profiles
• In-memory state
Simulation Clock Controller
• +5 / +15 / +30 min
• Auto-advance
• State management
Sensor Fault Injector
• Stale telemetry
• Packet dropouts
• Compression artifacts
• Missing / delayed data
3 Historical Feature Pipeline
CGM Trends
• 15m, 30m, 60m ROC
• Acceleration
Glycemic Variability
• Mean, min, max, std
• Rolling windows
Time in Range
• % in 70–180 mg/dL
• (previous 2 hours)
Wearable Features
• Heart rate trends
• Step count summaries
Circadian & EHR Features
• Time-of-day encoding
• Age, diabetes type, HbA1c, BMI
Feature window: previous 120 minutes through the current timestamp. Future readings are used only to construct training labels, never as inference features.
Offline Training (separate flow)
Training Data
(10 patients, 19,920 samples)
Model Training & Evaluation
Saved Model Artifact
glucotwin_model.joblib
+ metrics.json, pipeline_metadata.json
4 Machine Learning Model and Evaluation
Trained Model
• StandardScaler
• GradientBoostingClassifier
• Saved: glucotwin_model.joblib
Prediction Service
• Current feature vector
• Probability of excursion
• Threshold: 0.50
Held-Out Evaluation
• Train: 10 patients
• Test: PT-103 & PT-104
• ROC-AUC, Precision, Recall
• F1, Brier score
• Confusion matrix & class balance
Explainability
• Feature importance / attribution
• Contributions in dashboard
5 What-If Counterfactual Lab (Feedback Path)
Scenario Inputs
• Activity change
• Carbohydrate intake
• Sleep disruption
Synthetic Scenario Modification
• Modify relevant features
• Apply scenario rules
• Keep baseline & scenario separate
Model Re-scoring
• Use saved model
• Get scenario probability
Risk Comparison
• Baseline risk
• Scenario risk
• ΔRisk = P_scenario − P_baseline
Model-based hypothetical scenario, not a guaranteed physiological outcome.
6 FastAPI Backend
FastAPI
/api/patients
Patient & telemetry endpoints
/api/predict/:id
Prediction endpoint
/api/simulation
Simulation step, reset, anomaly endpoints
/api/whatif/simulate
What-If simulation endpoint
/api/metrics
Model metrics
7 React Clinical Dashboard
React
Clinical disclaimer & status header
Patient EHR profile
Current CGM & wearable vitals
Glucose time-series chart
Risk gauge
Feature explainability panel
What-If Lab
Telemetry quality monitor
Model evaluation panel
Legend:
Data Generation
Digital Twin / Simulation
Feature Engineering
ML Model
What-If (Feedback Path)
Backend API
Frontend
Main data flow
Training flow
Research prototype using synthetic data. Not clinically validated and not intended for diagnosis, treatment decisions, or medication recommendations. What-If outputs represent model-based hypothetical scenarios, not established causal effects.